

Univerza Alma Mater

**Projektna naloga pri predmetu
Računalništvo v oblaku in mobilne rešitve**

Namestitev Magento na AWS EC2

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Postopek namestitve Magento na AWS EC2

Za to projektno nalogo sem se odločil za namestitev sistema Magento, saj gre za eno najbolj razširjenih platform za razvoj spletnih trgovin. Magento omogoča enostavno upravljanje produktov, naročil, uporabnikov in ponuja veliko možnosti prilagoditve, zato je primeren tako za manjše kot tudi za večje spletne trgovine.

Za izvajanje projekta sem uporabil AWS EC2, ker omogoča hitro in fleksibilno postavitve virtualnega strežnika v oblaku. EC2 omogoča enostavno prilagajanje zmogljivosti (CPU, RAM), kar je še posebej pomembno pri zahtevnejših aplikacijah, kot je Magento.

Magento potrebuje več različnih komponent za delovanje, kot so spletni strežnik (Apache), podatkovna baza (MySQL) ter iskalni sistem (OpenSearch).

S tem projektom sem vzpostavil okolje v oblaku in omogočal delovanje in dostop do administracijskega vmesnika Magento.

1. Ustvarjanje EC2 instance

V AWS sem ustvaril novo EC2 instanco.

Operacijski sistem: Debian 13

Tip instance: t3.micro (kasneje spremenjeno na m7i-flex.large, zaradi potreb Magento servisa)

A VPC is an isolated portion of the AWS Cloud populated by AWS objects, such as Amazon EC2 instances.

Resources to create [Info](#)

Create only the VPC resource or the VPC and other networking resources.

Name tag - optional

Creates a tag with a key of 'Name' and a value that you specify.

IPv4 CIDR block [Info](#)

IPv4 CIDR

CIDR block size must be between /16 and /28.

IPv6 CIDR block [Info](#)

Tenancy [Info](#)

VPC encryption control (\$) | [Info](#)

☒ None

- ☐ **Enforce mode**
Requires all resources, except exclusions, in your VPC to be encryption-capable and blocks creation of unencrypted resources.

Create subnet [Info](#)

VPC ID

Create subnets in this VPC.

vpc-0596b925b349b4fcb (magentoVPC)

Associated VPC CIDRs

IPv4 CIDRs
10.0.0.0/16

Subnet settings

Specify the CIDR blocks and Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of 'Name' and a value that you specify.

private/Sub

The name can be up to 256 characters long.

Availability Zone [Info](#)
Choose the zone in which your subnet will reside, or let Amazon choose one for you.

Europe (Frankfurt) / eu-central-1 / eu-central-1b

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.

IPv4 subnet CIDR block

10.0.0.0/24 256 IP

▼ Tags - optional

Key Value - optional

[Add new tag](#)

Remove

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough Do not show me this message again

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name
magento-EC2

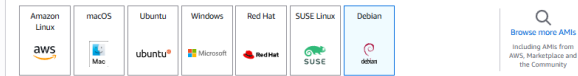
Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents Quick Start



Amazon Machine Image (AMI)
Debian 13 (HVM), SSD Volume Type
ami-0da1f66573556d917 (64-bit x86) / ami-08241d277446081d7 (64-bit ARM)
Virtualization: hvm - ENA: enabled: true - Root device type: ebs

Description
Debian 13 (HVM), EBS General Purpose (SSD) Volume Type. Community developed free GNU/Linux distribution. <https://www.debian.org/>
Debian 13 (20260316-2418)

Architecture: 64-bit (x86) AMI ID: ami-0da1f66573556d917 Publish Date: 2026-03-16 Username: admin

Instance type

Instance type
t3.micro
Family: t3 - 2 vCPU - 1 GiB Memory - Current generation: true - On-Demand SUSE base pricing: 0.012 USD per Hour - On-Demand RHEL base pricing: 0.0408 USD per Hour - On-Demand Ubuntu Pro base pricing: 0.0155 USD per Hour - On-Demand Linux base pricing: 0.012 USD per Hour - On-Demand Windows base pricing: 0.0212 USD per Hour

Summary

Number of instances: 1

Software Image (AMI)
Debian 13 (20260316-2418)
ami-0da1f66573556d917

Virtual server type (instance type)
t3.micro

Firewall (security group)
magentoSpletni

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

Instance type

Instance type

t3.micro
Family: t3 - 2 vCPU - 1 GiB Memory - Current generation: true - On-Demand SUSE base pricing: 0.012 USD per Hour - On-Demand RHEL base pricing: 0.0408 USD per Hour - On-Demand Ubuntu Pro base pricing: 0.0155 USD per Hour - On-Demand Linux base pricing: 0.012 USD per Hour - On-Demand Windows base pricing: 0.0212 USD per Hour

Additional costs apply for AMIs with pre-installed software

Key pair (login)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

Select Create new key pair

Network settings

VPC - required

vpc-0596b925b349b4fcb (magentoVPC)
10.0.0.0/16

Subnet

subnet-0a0e2d75c4520f822 javniSub
VPC: vpc-0596b925b349b4fcb Owner: 033313104891 Availability Zone: eu-central-1a (eu1-a22)
Zone type: Availability Zone - IP addresses available: 251 CIDR: 10.0.1.0/24

Auto-assign public IP

Enable

Firewall (security groups)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Create security group Select existing security group

Common security groups

Select security groups Compare security group rules

magentoSpletni sg-04ade166a80473e94
VPC: vpc-0596b925b349b4fcb

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Advanced network configuration

Configure storage

Advanced

Number of instances

1

Software Image (AMI)
Debian 13 (20260316-2418)
ami-0da1f66573556d917

Virtual server type (instance type)
t3.micro

Firewall (security group)
magentoSpletni

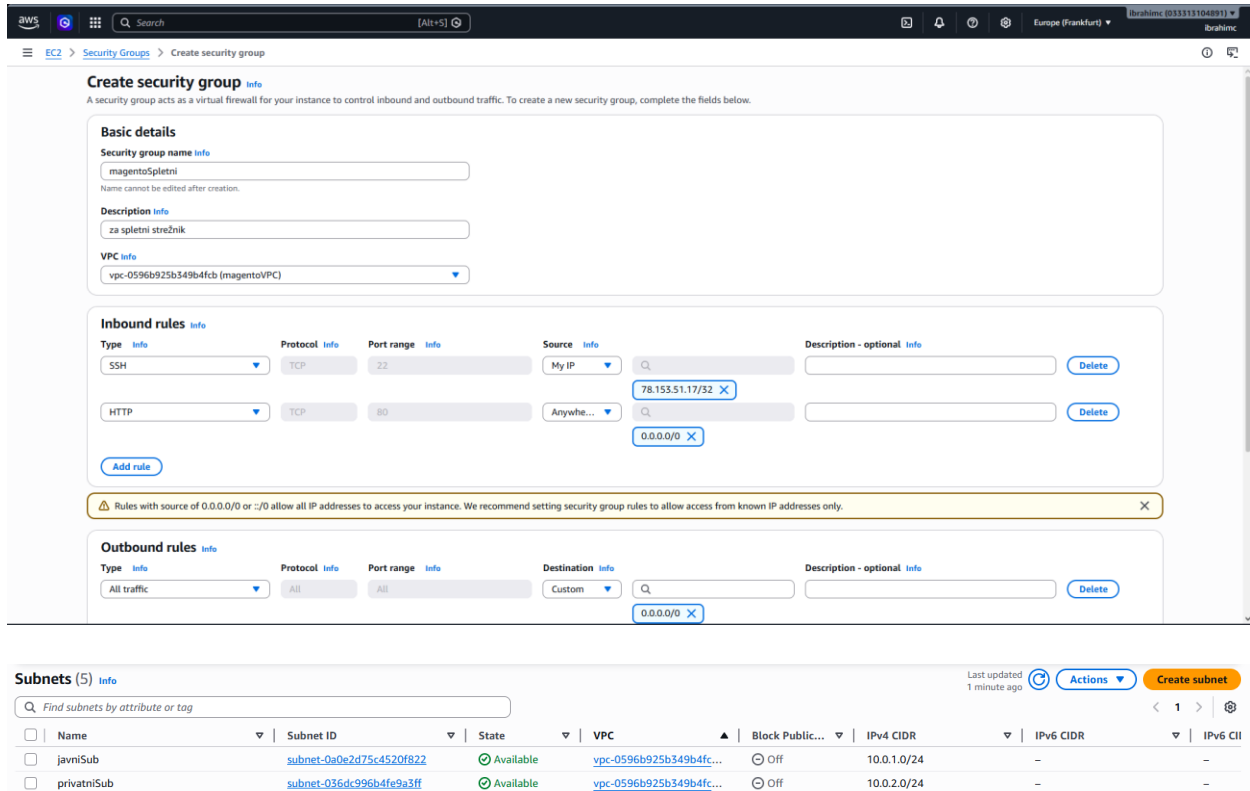
Storage (volumes)
1 volume(s) - 8 GiB

Cancel Launch instance Preview code

2. Nastavitev omrežja (Security Group in Subnet)

SSH (port 22) – samo moj IP

HTTP (port 80) – dostop za vse (0.0.0.0/0)



The screenshot displays the AWS Management Console interface. The top navigation bar shows the AWS logo, a search bar, and the user's profile. The main content area is titled 'Create security group' and includes a description: 'A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.'

Basic details

- Security group name:** magentoSpletni
- Description:** za spletni strežnik
- VPC:** vpc-0596b925b349b4fc (magentoVPC)

Inbound rules

Type	Protocol	Port range	Source	Description - optional	Actions
SSH	TCP	22	My IP		Delete
HTTP	TCP	80	Anywhere...		Delete

Outbound rules

Type	Protocol	Port range	Destination	Description - optional	Actions
All traffic	All	All	Custom		Delete

Subnets (5)

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR	IPv6 CIDR	IPv6 CLI
javniSub	subnet-0a0e2d75c4520f822	Available	vpc-0596b925b349b4fc...	Off	10.0.1.0/24	-	-
privatniSub	subnet-036dc996b4fe9a3ff	Available	vpc-0596b925b349b4fc...	Off	10.0.2.0/24	-	-

3. Namestitev Composerja

Composer je potreben za namestitev Magento. Preveril sem če je nameščen in dostopen.

```

admin@ip-10-0-1-9:~$ curl -sS https://getcomposer.org/installer | php
All settings correct for using Composer
Downloading...

Composer (version 2.9.7) successfully installed to: /home/admin/composer.phar
Use it: php composer.phar

admin@ip-10-0-1-9:~$ sudo mv composer.phar /usr/local/bin/composer
admin@ip-10-0-1-9:~$ composer --version
Composer version 2.9.7 2026-04-14 13:31:52
PHP version 8.4.16 (/usr/bin/php8.4)
Run the "diagnose" command to get more detailed diagnostics output.
admin@ip-10-0-1-9:~$

```

```

admin@ip-10-0-1-9:/var/www/magento$ composer create-project --repository=https://repo.magento.com/ magento/project-community-edition .
Warning from repo.magento.com: You haven't provided your Magento Authentication Keys. For instructions, visit https://devdocs.magento.com/guides/v2.3/install-gde/prereq/connect-auth.html
Authentication required (repo.magento.com):
Username: 940cf22a4db0e0b413f39887645d5204
Password:
Warning from repo.magento.com: Your Magento authentication keys are invalid. Please double-check your keys in your Marketplace account. For instructions, visit https://devdocs.magento.com/guides/v2.3/install-gde/prereq/connect-auth.html
In AuthHelper.php line 229:

  Invalid credentials (HTTP 401) for 'https://repo.magento.com/packages.json', aborting.

create-project [-d|--directory DIRECTORY] [--prefer-dist] [--prefer-install PREFER-INSTALL] [--repository REPOSITORY] [--repository-url REPOSITORY-URL] [--add-repository] [--dev] [--no-dev] [--no-custom-installers] [--no-scripts] [--no-progress] [--no-secure-http] [--keep-vcs] [--remove-vcs] [--no-install] [--no-audit] [--audit-format AUDIT-FORMAT] [--no-security-blocking] [--ignore-platform-req IGNORE-PLATFORM-REQ] [--ignore-platform-reqs] [--ask] [--]
] [] [] []

admin@ip-10-0-1-9:/var/www/magento$ composer config --global http-basic.repo.magento.com 940cf22a4db0e0b413f39887645d5204 d81e45d4177fefe43a320eaf347b9
admin@ip-10-0-1-9:/var/www/magento$ composer create-project --repository=https://repo.magento.com/ magento/project-community-edition .
Creating a "magento/project-community-edition" project at "."
Installing magento/project-community-edition (2.4.8-p4)
- Downloading magento/project-community-edition (2.4.8-p4)
- Installing magento/project-community-edition (2.4.8-p4): Extracting archive
Created project in /var/www/magento/.
Loading composer repositories with package information
Updating dependencies
Lock file operations: 628 installs, 0 updates, 0 removals
- Locking livewire/livewire (1.0.2)
- Locking adobe/commerce/adobe-ae-metapackage (2.2.3)
- Locking adobe-commerce/ae-extensions-metapackage (1.0.2)
- Locking allure-framework/allure-codeception (v2.6.0)
- Locking allure-framework/allure-symfony (v2.4.0)
- Locking arach/arach-api-libphp (1.1.7)
- Locking awa/awa-ctrl-php (v1.2.7)
- Locking awa/awa-sb-ctrl (1.37.11)
- Locking bacon/bacon-qr-code (v3.1.1)
- Locking behat/gherkin (v4.14.1)
- Locking braintree/braintree_php (6.21.0)
- Locking brich/match (0.12.3)
- Locking brich/magento2 (0.6.0)
- Locking carlos-matos/match (0.0.17)
- Locking christian-riesen/based (1.6.0)
- Locking classmate/swext (v1.3.0)
- Locking codeception/codeception (5.3.5)
- Locking codeception/lib-asserts (2.2.0)
- Locking codeception/lib-asserts (1.0.7)
- Locking codeception/module-asserts (3.2.1)
- Locking codeception/module-webdriver (4.0.5)
- Locking codeception/php (6.3.0)
- Locking colasmolleshou/cache-backend-file (v1.4.8)
- Locking colasmolleshou/cache-backend-pedis (1.18.0)
- Locking colasmolleshou/redis (v1.17.1)

```

4. Namestitev PHP in Apache

Potem sem namestil PHP in Apache strežnik, omogočil potrebne module in ponovno zagnal Apache.

```

admin@ip-10-0-1-9:~$ php -v
PHP 8.4.16 (cli) (built: Dec 18 2025 21:19:25) (NTS)
Copyright (c) The PHP Group
Built by Debian
Zend Engine v4.4.16, Copyright (c) Zend Technologies
    with Zend OPcache v8.4.16, Copyright (c), by Zend Technologies
admin@ip-10-0-1-9:~$ sudo a2enmod php8.2
ERROR: Module php8.2 does not exist!
admin@ip-10-0-1-9:~$ sudo a2enmod php8.4
Considering dependency mpm_prefork for php8.4:
Considering conflict mpm_event for mpm_prefork:
Considering conflict mpm_worker for mpm_prefork:
Module mpm_prefork already enabled
Considering conflict php5 for php8.4:
Module php8.4 already enabled
admin@ip-10-0-1-9:~$ sudo systemctl restart apache2

```

5. Namestitev MySQL klienta in povezava na RDS

```

admin@ip-10-0-1-9:/var/www/magento$ sudo apt install default-mysql-client
Installing:
  default-mysql-client

Installing dependencies:
  libconfig-inifiles-perl libdbd-mariadb-perl libdbi-perl libmariadb3 libncurses6 libpcre2-posix3 libterm-readkey-perl mariadb-client mariadb-client-compat mariadb-client-core mariadb-common mysql-common

Suggested packages:
  libclone-perl libmaildm-perl libnet-daemon-perl libsql-statement-perl

Summary:
  Upgrading: 0, Installing: 13, Removing: 0, Not Upgrading: 0
  Download size: 5531 kB
  Space needed: 86.1 MB / 5354 MB available

Continue? [Y/n] y
Get:1 file:/etc/apt/mirrors/debian.11st Mirrorlist [39 B]
Get:2 https://cdn-aws.deb.debian.org/debian trixie/main amd64 libconfig-inifiles-perl all 3.000003-3 [44.8 kB]
Get:3 https://cdn-aws.deb.debian.org/debian trixie/main amd64 mysql-common all 8.0+1.1.1 [6784 B]
Get:4 https://cdn-aws.deb.debian.org/debian trixie/main amd64 mariadb-common all 1:11.8.6-0+deb13u1 [29.5 kB]
Get:5 https://cdn-aws.deb.debian.org/debian trixie/main amd64 libmariadb3 amd64 1:11.8.6-0+deb13u1 [187 kB]
Get:6 https://cdn-aws.deb.debian.org/debian trixie/main amd64 libncurses6 amd64 6.5+20250216-2 [105 kB]
Get:7 https://cdn-aws.deb.debian.org/debian trixie/main amd64 mariadb-client-core amd64 1:11.8.6-0+deb13u1 [919 kB]
Get:8 https://cdn-aws.deb.debian.org/debian trixie/main amd64 libpcre2-posix3 amd64 10.46-1-deb13u1 [63.9 kB]
Get:9 https://cdn-aws.deb.debian.org/debian trixie/main amd64 mariadb-client amd64 1:11.8.6-0+deb13u1 [3164 kB]
Get:10 https://cdn-aws.deb.debian.org/debian trixie/main amd64 mariadb-client-compat all 1:11.8.6-0+deb13u1 [29.7 kB]
Get:11 https://cdn-aws.deb.debian.org/debian trixie/main amd64 default-mysql-client all 1:1.1.1 [3028 B]
Get:12 https://cdn-aws.deb.debian.org/debian trixie/main amd64 libdbi-perl amd64 1.647-1 [861 kB]
Get:13 https://cdn-aws.deb.debian.org/debian trixie/main amd64 libdbd-mariadb-perl amd64 1.22-1+deb4 [93.6 kB]
Get:14 https://cdn-aws.deb.debian.org/debian trixie/main amd64 libterm-readkey-perl amd64 2.38-2+b4 [24.6 kB]
Fetched 5531 kB in 0s (38.0 MB/s)
Selecting previously unselected package libconfig-inifiles-perl.
(Reading database ... 35636 files and directories currently installed.)
Preparing to unpack .../00-libconfig-inifiles-perl_3.000003-3_all.deb ...

```

```

admin@ip-10-0-1-9:/var/www/magento$ mysql -h magento-rds.cteqq0ge0wbv.eu-central-1.rds.amazonaws.com \
-u admin -p \
--ssl-ca=global-bundle.pem
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 40
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> 

```

```

MySQL [(none)]> SELECT user, host FROM mysql.user WHERE user='magento';
+-----+-----+
| user      | host      |
+-----+-----+
| magento   | %         |
+-----+-----+
1 row in set (0.015 sec)

MySQL [(none)]> exit;
Bye
admin@ip-10-0-1-9:/var/www/magento$ mysql -h magento-rds.cteqq0ge0wbw.eu-central-1.rds.amazonaws.com -u magento -p --skip-ssl magento
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 52
Server version: 8.4.8 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [magento]> exit;
Bye

```

6. Namestitev OpenSearch

Namestil sem OpenSearch, ki ga Magento uporablja za iskanje. Imel sem težavo z zagonom zaradi pomnilnika, kar sem rešil z nadgradnjo instance.

```
wget https://artifacts.opensearch.org/releases/bundle/opensearch/3.2.0/opensearch-3.2.0-linux-x64.deb
```

```
sudo dpkg -i opensearch-3.2.0-linux-x64.deb
```

```

admin@ip-10-0-1-9:/var/www/magento$ wget https://artifacts.opensearch.org/releases/bundle/opensearch/3.2.0/opensearch-3.2.0-linux-x64.deb
--2026-05-04 15:02:33-- https://artifacts.opensearch.org/releases/bundle/opensearch/3.2.0/opensearch-3.2.0-linux-x64.deb
Resolving artifacts.opensearch.org (artifacts.opensearch.org)... 52.222.236.4, 52.222.236.105, 52.222.236.15, ...
Connecting to artifacts.opensearch.org (artifacts.opensearch.org)... connected.
HTTP request sent, awaiting response... 200 OK
Length: 939006822 (890M) [application/octet-stream]
Saving to: 'opensearch-3.2.0-linux-x64.deb'

opensearch-3.2.0-linux-x64.deb
100%[=====] 895.51M 18.9MB/s in 51s

2026-05-04 15:02:33 (17.7 MB/s) - 'opensearch-3.2.0-linux-x64.deb' saved [939006822/939006822]

admin@ip-10-0-1-9:/var/www/magento$ sudo env OPENSEARCH_INITIAL_ADMIN_PASSWORD='Admin123456!' dpkg -i opensearch-3.2.0-linux-x64.deb
Selecting previously unselected package opensearch.
(Reading database ... 36000 files and directories currently installed.)
Preparing to unpack opensearch-3.2.0-linux-x64.deb ...
Running opensearch Post-Installation Script
Unpacking opensearch (3.2.0) ...
Setting up opensearch (3.2.0) ...
Running Opensearch Post-Installation Script

```

7. Nastavitev pravic(Permissions)

Nastavil sem ustrezne pravice nad mapami Magento, da lahko sistem pravilno zapisuje podatke.

```

admin@ip-10-0-1-9:/var/www/magento$ sudo find var generated vendor pub/static pub/media app/etc -type d -exec chmod g+ws {} +
admin@ip-10-0-1-9:/var/www/magento$ sudo chown -R :www-data .
admin@ip-10-0-1-9:/var/www/magento$ sudo chmod u+x bin/magento
admin@ip-10-0-1-9:/var/www/magento$

```

8. Namestitev Magento

Magento sem namestil z uporabo Composerja.

```
Module 'Magento_Usp':
[Progress: 1449 / 1476]
Module 'Magento_InventoryConfigurableProductFrontendUi':
[Progress: 1450 / 1476]
Module 'Magento_PaypalCaptcha':
[Progress: 1451 / 1476]
Module 'Magento_VaultGraphQL':
[Progress: 1452 / 1476]
Module 'Magento_Version':
[Progress: 1453 / 1476]
Module 'Magento_InventoryInStorePickupWebapiExtension':
[Progress: 1454 / 1476]
Module 'Magento_WebapiAsync':
[Progress: 1455 / 1476]
Module 'Magento_WebapiSecurity':
[Progress: 1456 / 1476]
Module 'Magento_Weee':
[Progress: 1457 / 1476]
Module 'Magento_WeeeGraphQL':
[Progress: 1458 / 1476]
Module 'Magento_CurrencySymbol':
[Progress: 1459 / 1476]
Module 'Magento_Wishlist':
[Progress: 1460 / 1476]
Module 'Magento_WishlistAnalytics':
[Progress: 1461 / 1476]
Module 'Magento_WishlistGraphQL':
[Progress: 1462 / 1476]
Module 'PayPal_Braintree':
[Progress: 1463 / 1476]
Module 'PayPal_BraintreeCustomerBalance':
[Progress: 1464 / 1476]
Module 'PayPal_BraintreeGiftCard':
[Progress: 1465 / 1476]
Module 'PayPal_BraintreeGiftCardAccount':
[Progress: 1466 / 1476]
Module 'PayPal_BraintreeGiftWrapping':
[Progress: 1467 / 1476]
Module 'PayPal_BraintreeGraphQL':
[Progress: 1468 / 1476]
Module 'PayPal_BraintreeReward':
[Progress: 1469 / 1476]
[Progress: 1470 / 1476]
Installing admin user...
[Progress: 1471 / 1476]
Caches clearing:
[SUCCESS]: Cache cleared successfully
[Progress: 1472 / 1476]
Disabling Maintenance Mode:
[Progress: 1473 / 1476]
Post installation file permissions check...
For security, remove write permissions from these directories: '/var/www/magento/app/etc'
[Progress: 1474 / 1476]
Write installation date...
[Progress: 1475 / 1476]
Indexing...
14 indexer(s) are indexed.
[Progress: 1476 / 1476]
[SUCCESS]: Magento installation complete.
[SUCCESS]: Magento Admin URI: /admin_otpztqm
Nothing to import.
```

Spremenil sem public IP zaradi spremenjenega tipa instance:

```
admin@ip-10-0-1-9:/var/www/magento$ php bin/magento setup:store-config:set --base-
url="http://3.72.21.13/"
```

```
admin@ip-10-0-1-9:/var/www/magento$ php bin/magento cache:flush
```

9. Nastavitev Apache virtual host-a

```
sudo nano /etc/apache2/sites-available/magento.conf
```

```
sudo a2ensite magento.conf
```

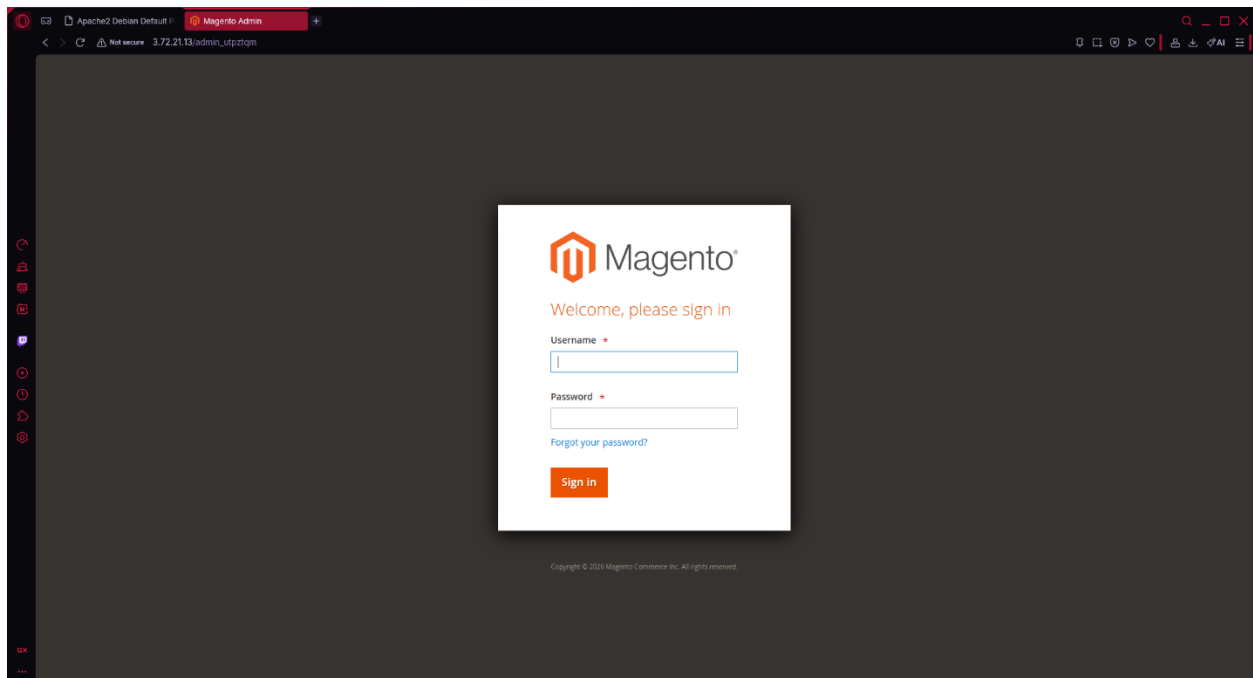
```
sudo a2enmod rewrite
```

```
sudo systemctl restart apache2
```

```
No security vulnerability advisories found.
admin@ip-10-0-1-9:/var/www/magento$ sudo nano /etc/apache2/sites-available/magento.conf
admin@ip-10-0-1-9:/var/www/magento$ sudo a2ensite magento.conf
Enabling site magento.
To activate the new configuration, you need to run:
    systemctl reload apache2
admin@ip-10-0-1-9:/var/www/magento$ sudo a2enmod rewrite
Enabling module rewrite.
To activate the new configuration, you need to run:
    systemctl restart apache2
admin@ip-10-0-1-9:/var/www/magento$ sudo systemctl restart apache2
admin@ip-10-0-1-9:/var/www/magento$
```

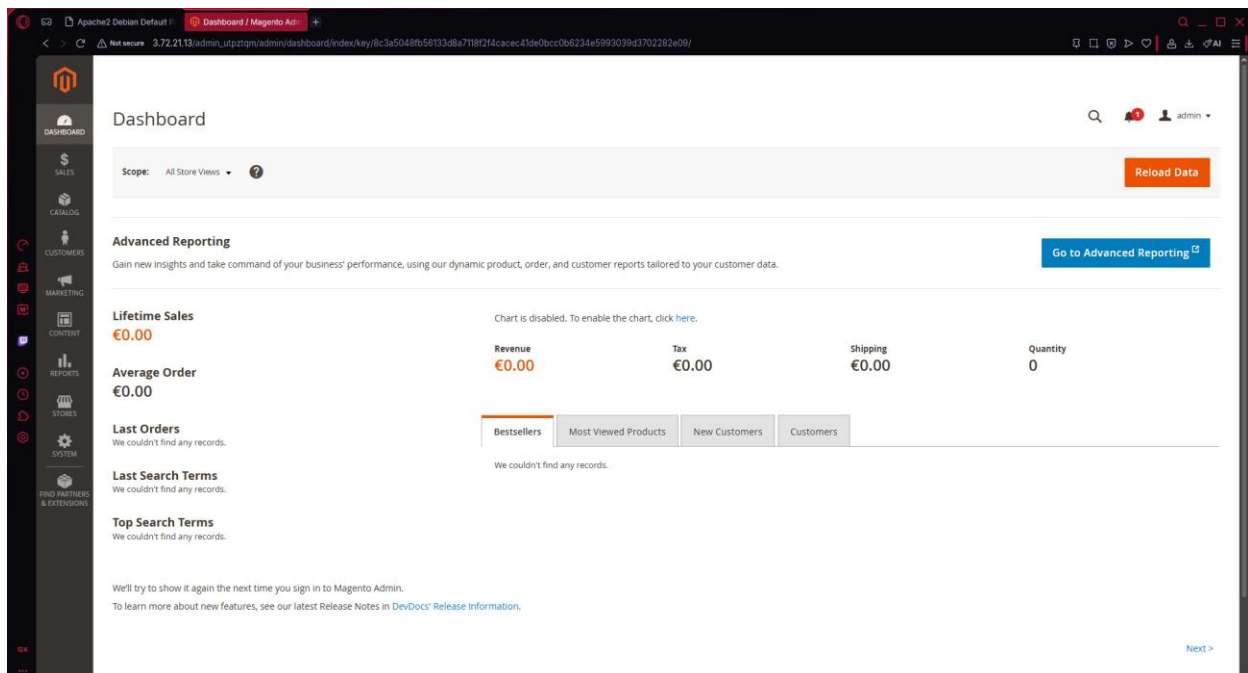
10. Dostop do Magento admin panela:

Preko brskalnika sem dostopal do Magento admin panela.



11. Uspešna prijava v Magento

Na koncu sem se uspešno prijavil v Magento admin.



Viri:

<https://commercemarketplace.adobe.com>

<https://aws.amazon.com/free/>